

Pyridine Phosphination - Binary - Model Testing (10% holdout)

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Model Search

formula	McFadden R2
class ~ fr_Ar_N + fr_aryl_methyl + fr_benzene + fr_halogen	0.721
class ~ fr_aryl_methyl + fr_benzene + fr_halogen + fr_pyridine	0.721
class ~ NPA_1_P + fr_Ar_N + fr_aryl_methyl + fr_benzene	0.689
class ~ NPA_1_P + B1_2_SM + Iso.Polar_SM + fr_Ar_N	0.683
class ~ cross_P + fr_Ar_N + fr_aryl_methyl + fr_halogen	0.676
class ~ cross_P + fr_aryl_methyl + fr_halogen + fr_pyridine	0.676
class ~ NPA_1_P + Iso.Polar_SM + fr_Ar_N + fr_ether	0.667
class ~ fr_Ar_N + fr_aryl_methyl + fr_benzene + fr_ether	0.662
class ~ NPA_5_P + fr_Ar_N + fr_aryl_methyl + fr_benzene	0.657
class ~ NPA_Avg_H_onPhos + fr_Ar_N + fr_aryl_methyl + fr_benzene	0.644

Section Results Summary

Model	Training Accuracy	Test Accuracy
1. 1st Place	88.37	64.29
2. 2nd Place	72.09	71.43
3. 3rd Place	93.02	64.29
4. 4th Place	90.70	64.29
5. 5th Place	83.72	50.00
6. 6th Place	69.77	71.43
7. 7th Place	90.70	64.29
8. 8th Place	88.37	64.29
9. 9th Place	90.70	64.29
10. 10th Place	93.02	64.29

1st Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
88.37%	0.572

Model Coefficients

Term	Estimate
(Intercept)	11.8471427
fr_Ar_N	-10.7659931
fr_aryl_methyl	-3.2318692
fr_benzene	-3.1569813
fr_halogen	0.4566335

Training Set

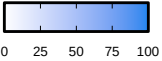
1	28 (90%)	3 (10%)	31 (72%)
2	2 (17%)	10 (83%)	12 (28%)
total	30 (93%)	13 (77%)	43 (88%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	100	0	1
AF	84	16	1
B	89	11	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	89	11	1
AD	98	2	1
AE	100	0	1
AH	84	16	1
AI	100	0	1
AL	99	1	1
AM	90	10	1
H	25	75	1
K	69	31	1
N	100	0	1
O	89	11	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	93	7	1
V	89	11	1
W	99	1	1
X	100	0	1
Y	99	1	1
Z	99	1	1
AN	45	55	1
AO	25	75	1
AP	89	11	1
AS	99	1	1
AV	25	75	2
AW	25	75	2
AX	84	16	2
AY	8	92	2
AZ	25	75	2
BA	8	92	2
BB	18	82	2
BC	25	75	2
BE	0	100	2
BF	0	100	2
BH	45	55	2
BI	83	17	2



Test Set

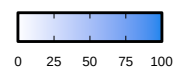
1	7 (70%)	3 (30%)	10 (71%)
2	2 (50%)	2 (50%)	4 (29%)
total	9 (78%)	5 (40%)	14 (64%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set

Probability Heatmap

	1	2	Exp
AA	25	75	1
AB	25	75	1
AJ	100	0	1
AK	90	10	1
G	25	75	1
I	83	17	1
M	89	11	1
AQ	83	17	1
AT	100	0	1
AU	89	11	1
BD	25	75	2
BG	76	24	2
BJ	18	82	2
BK	89	11	2



1. 1st Place

2nd Place

Table:

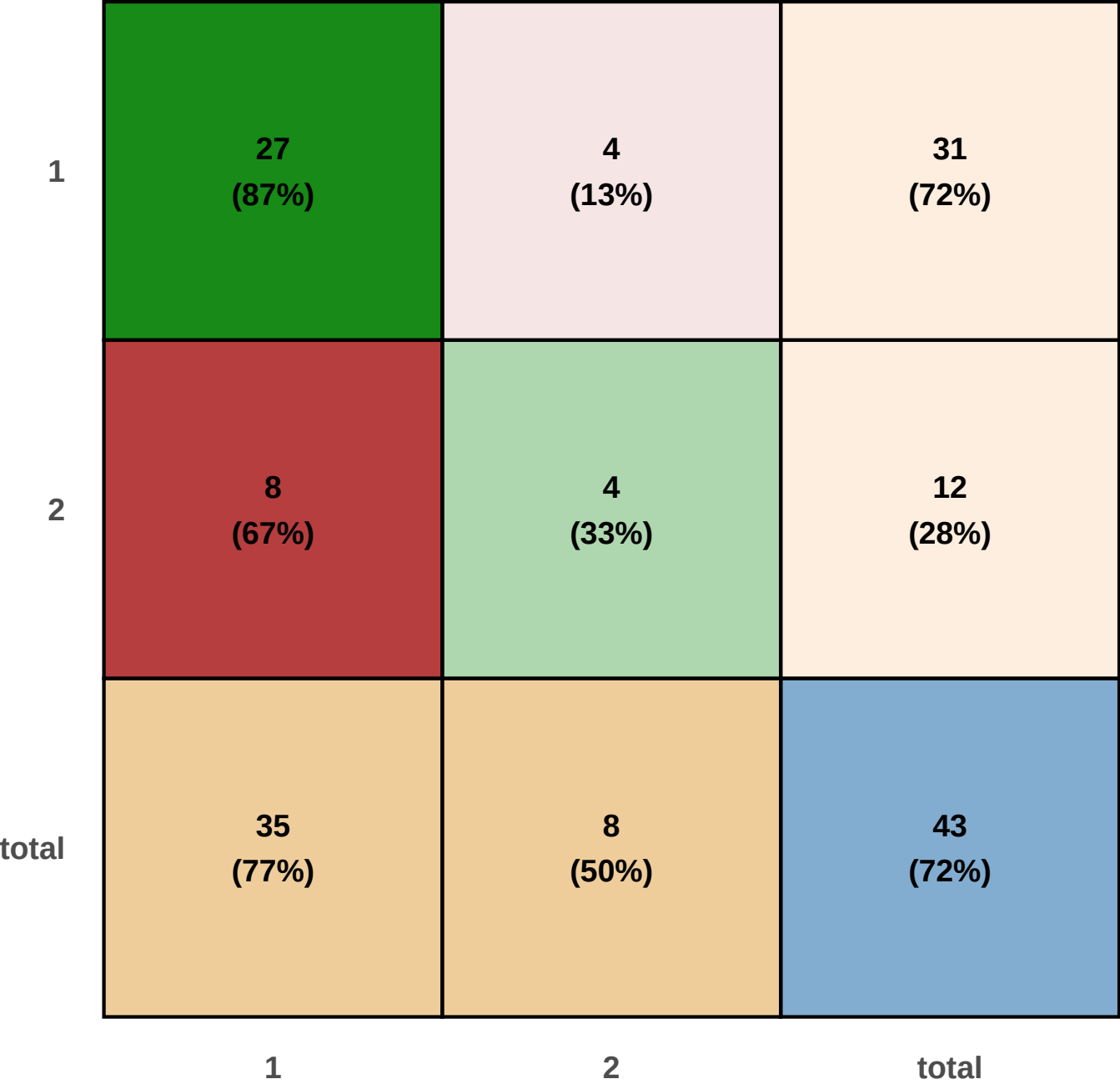
Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
72.09%	0.239

Model Coefficients

Term	Estimate
(Intercept)	0.4523818
fr_aryl_methyl	-2.2346386
fr_benzene	-2.0872847
fr_halogen	0.4129210
fr_pyridine	-0.6057780

Training Set

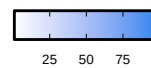


■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	99	1	1
AF	88	12	1
B	90	10	1
D	50	50	1
E	99	1	1
F	72	28	1
AC	90	10	1
AD	96	4	1
AE	84	16	1
AH	88	12	1
AI	68	32	1
AL	97	3	1
AM	92	8	1
H	54	46	1
K	76	24	1
N	39	61	1
O	90	10	1
P	68	32	1
R	92	8	1
S	39	61	1
T	39	61	1
U	88	12	1
V	90	10	1
W	98	2	1
X	77	23	1
Y	98	2	1
Z	98	2	1
AN	54	46	1
AO	54	46	1
AP	90	10	1
AS	97	3	1
AV	54	46	2
AW	54	46	2
AX	88	12	2
AY	25	75	2
AZ	54	46	2
BA	25	75	2
BB	44	56	2
BC	54	46	2
BE	30	70	2
BF	77	23	2
BH	54	46	2
BI	86	14	2



2. 2nd Place

Test Set

1	9 (90%)	1 (10%)	10 (71%)
2	3 (75%)	1 (25%)	4 (29%)
total	12 (75%)	2 (50%)	14 (71%)
	1	2	total

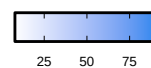
■ Accuracy ■ False ■ Precision ■ Size ■ True

2. 2nd Place

Test Set

Probability Heatmap

	1	2	Exp
AA	54	46	1
AB	54	46	1
AJ	38	62	1
AK	92	8	1
G	54	46	1
I	86	14	1
M	90	10	1
AQ	86	14	1
AT	72	28	1
AU	90	10	1
BD	54	46	2
BG	80	20	2
BJ	44	56	2
BK	90	10	2



2. 2nd Place

3rd Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
93.02%	0.689

Model Coefficients

Term	Estimate
(Intercept)	4.018504
NPA_1_P	103.474680
fr_Ar_N	-90.226358
fr_aryl_methyl	-3.407754
fr_benzene	-6.435189

Training Set

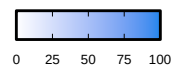
1	29 (94%)	2 (6%)	31 (72%)
2	1 (8%)	11 (92%)	12 (28%)
total	30 (97%)	13 (85%)	43 (93%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	100	0	1
AF	97	3	1
B	100	0	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	98	2	1
AD	100	0	1
AE	100	0	1
AH	87	13	1
AI	100	0	1
AL	100	0	1
AM	76	24	1
H	32	68	1
K	80	20	1
N	100	0	1
O	100	0	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	100	0	1
V	99	1	1
W	91	9	1
X	100	0	1
Y	90	10	1
Z	100	0	1
AN	99	1	1
AO	10	90	1
AP	98	2	1
AS	100	0	1
AV	2	98	2
AW	31	69	2
AX	80	20	2
AY	34	66	2
AZ	8	92	2
BA	21	79	2
BB	13	87	2
BC	29	71	2
BE	0	100	2
BF	0	100	2
BH	9	91	2
BI	17	83	2



Test Set

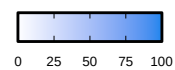
1	7 (70%)	3 (30%)	10 (71%)
2	2 (50%)	2 (50%)	4 (29%)
total	9 (78%)	5 (40%)	14 (64%)
	1	2	total

■ Accuracy
■ False
■ Precision
■ Size
■ True

Test Set

Probability Heatmap

	1	2	Exp
AA	19	81	1
AB	6	94	1
AJ	100	0	1
AK	83	17	1
G	0	100	1
I	99	1	1
M	99	1	1
AQ	69	31	1
AT	100	0	1
AU	98	2	1
BD	5	95	2
BG	98	2	2
BJ	48	52	2
BK	99	1	2



4th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
90.7%	0.683

Model Coefficients

Term	Estimate
:-----:	-----:
(Intercept)	-81.6029643
NPA_1_P	201.4899686
B1_2_SM	7.0023907
Iso.Polar_SM	-0.1047517
fr_Ar_N	-92.8475876

Training Set

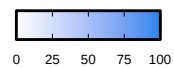
1	28 (90%)	3 (10%)	31 (72%)
2	1 (8%)	11 (92%)	12 (28%)
total	29 (97%)	14 (79%)	43 (91%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	100	0	1
AF	97	3	1
B	100	0	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	100	0	1
AD	100	0	1
AE	100	0	1
AH	45	55	1
AI	100	0	1
AL	96	4	1
AM	72	28	1
H	82	18	1
K	24	76	1
N	100	0	1
O	100	0	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	100	0	1
V	86	14	1
W	100	0	1
X	100	0	1
Y	98	2	1
Z	100	0	1
AN	100	0	1
AO	32	68	1
AP	100	0	1
AS	100	0	1
AV	12	88	2
AW	82	18	2
AX	49	51	2
AY	32	68	2
AZ	8	92	2
BA	25	75	2
BB	3	97	2
BC	31	69	2
BE	0	100	2
BF	0	100	2
BH	4	96	2
BI	21	79	2



Test Set

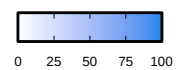
1	8 (80%)	2 (20%)	10 (71%)
2	3 (75%)	1 (25%)	4 (29%)
total	11 (73%)	3 (33%)	14 (64%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set

Probability Heatmap

	1	2	Exp
AA	100	0	1
AB	100	0	1
AJ	100	0	1
AK	26	74	1
G	2	98	1
I	98	2	1
M	100	0	1
AQ	97	3	1
AT	100	0	1
AU	94	6	1
BD	29	71	2
BG	97	3	2
BJ	100	0	2
BK	100	0	2



5th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
83.72%	0.508

Model Coefficients

Term	Estimate
:-----:	-----:
(Intercept)	-143.5908649
cross_P	0.1809144
fr_Ar_N	-153.0674365
fr_aryl_methyl	-4.3359768
fr_halogen	0.1499028

Training Set

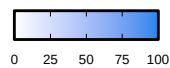
1	28 (90%)	3 (10%)	31 (72%)
2	4 (33%)	8 (67%)	12 (28%)
total	32 (88%)	11 (73%)	43 (84%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	98	2	1
AF	100	0	1
B	81	19	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	56	44	1
AD	100	0	1
AE	100	0	1
AH	100	0	1
AI	100	0	1
AL	89	11	1
AM	99	1	1
H	62	38	1
K	89	11	1
N	100	0	1
O	95	5	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	75	25	1
V	47	53	1
W	98	2	1
X	100	0	1
Y	71	29	1
Z	100	0	1
AN	34	66	1
AO	32	68	1
AP	53	47	1
AS	92	8	1
AV	38	62	2
AW	62	38	2
AX	81	19	2
AY	60	40	2
AZ	8	92	2
BA	65	35	2
BB	13	87	2
BC	45	55	2
BE	0	100	2
BF	0	100	2
BH	16	84	2
BI	42	58	2



Test Set

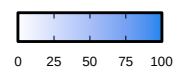
1	6 (60%)	4 (40%)	10 (71%)
2	3 (75%)	1 (25%)	4 (29%)
total	9 (67%)	5 (20%)	14 (50%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set

Probability Heatmap

	1	2	Exp
AA	41	59	1
AB	59	41	1
AJ	100	0	1
AK	96	4	1
G	26	74	1
I	58	42	1
M	18	82	1
AQ	11	89	1
AT	100	0	1
AU	62	38	1
BD	19	81	2
BG	100	0	2
BJ	93	7	2
BK	65	35	2



5. 5th Place

6th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
69.77%	0.11

Model Coefficients

Term	Estimate
(Intercept)	-8.2072850
cross_P	0.0048379
fr_aryl_methyl	-1.8492228
fr_halogen	0.1090619
fr_pyridine	-0.4294586

Training Set

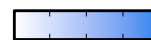
1	30 (97%)	1 (3%)	31 (72%)
2	12 (100%)	0 (0%)	12 (28%)
total	42 (71%)	1 (0%)	43 (70%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	99	1	1
AF	92	8	1
B	68	32	1
D	44	56	1
E	99	1	1
F	83	17	1
AC	67	33	1
AD	64	36	1
AE	64	36	1
AH	92	8	1
AI	76	24	1
AL	91	9	1
AM	93	7	1
H	67	33	1
K	90	10	1
N	59	41	1
O	69	31	1
P	73	27	1
R	92	8	1
S	50	50	1
T	52	48	1
U	86	14	1
V	67	33	1
W	92	8	1
X	67	33	1
Y	91	9	1
Z	93	7	1
AN	54	46	1
AO	66	34	1
AP	67	33	1
AS	64	36	1
AV	67	33	2
AW	67	33	2
AX	91	9	2
AY	60	40	2
AZ	65	35	2
BA	60	40	2
BB	63	37	2
BC	67	33	2
BE	50	50	2
BF	61	39	2
BH	53	47	2
BI	64	36	2



25 50 75

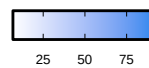
Test Set

1	10 (100%)	0 (0%)	10 (71%)
2	4 (100%)	0 (0%)	4 (29%)
total	14 (71%)	0 (0%)	14 (71%)
	1	2	total

■ Accuracy
■ False
■ Precision
■ Size
■ True

Test Set
Probability Heatmap

	1	2	Exp
AA	67	33	1
AB	67	33	1
AJ	67	33	1
AK	93	7	1
G	66	34	1
I	65	35	1
M	66	34	1
AQ	63	37	1
AT	81	19	1
AU	67	33	1
BD	66	34	2
BG	66	34	2
BJ	66	34	2
BK	67	33	2



7th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
90.7%	0.606

Model Coefficients

Term	Estimate
:-----:	-----:
(Intercept)	-34.7609732
NPA_1_P	150.1299225
Iso.Polar_SM	-0.0664514
fr_Ar_N	-85.4211041
fr_ether	-1.1263956

Training Set

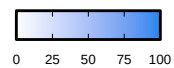
1	29 (94%)	2 (6%)	31 (72%)
2	2 (17%)	10 (83%)	12 (28%)
total	31 (94%)	12 (83%)	43 (91%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	100	0	1
AF	79	21	1
B	99	1	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	92	8	1
AD	100	0	1
AE	100	0	1
AH	29	71	1
AI	100	0	1
AL	96	4	1
AM	69	31	1
H	45	55	1
K	55	45	1
N	100	0	1
O	100	0	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	100	0	1
V	72	28	1
W	98	2	1
X	100	0	1
Y	80	20	1
Z	100	0	1
AN	90	10	1
AO	67	33	1
AP	100	0	1
AS	100	0	1
AV	41	59	2
AW	45	55	2
AX	76	24	2
AY	25	75	2
AZ	10	90	2
BA	20	80	2
BB	2	98	2
BC	89	11	2
BE	0	100	2
BF	0	100	2
BH	1	99	2
BI	19	81	2



Test Set

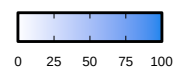
1	8 (80%)	2 (20%)	10 (71%)
2	3 (75%)	1 (25%)	4 (29%)
total	11 (73%)	3 (33%)	14 (64%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set

Probability Heatmap

	1	2	Exp
AA	99	1	1
AB	100	0	1
AJ	100	0	1
AK	19	81	1
G	1	99	1
I	89	11	1
M	99	1	1
AQ	100	0	1
AT	100	0	1
AU	55	45	1
BD	17	83	2
BG	76	24	2
BJ	100	0	2
BK	100	0	2



8th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
88.37%	0.536

Model Coefficients

Term	Estimate
(Intercept)	13.2799568
fr_Ar_N	-11.7180472
fr_aryl_methyl	-3.0468217
fr_benzene	-2.6807445
fr_ether	-0.4880699

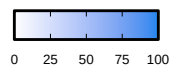
Training Set

1	29 (94%)	2 (6%)	31 (72%)
2	3 (25%)	9 (75%)	12 (28%)
total	32 (91%)	11 (82%)	43 (88%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set
Probability Heatmap

	1	2	Exp
A	99	1	1
AF	82	18	1
B	75	25	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	75	25	1
AD	99	1	1
AE	100	0	1
AH	82	18	1
AI	100	0	1
AL	98	2	1
AM	82	18	1
H	17	83	1
K	82	18	1
N	100	0	1
O	75	25	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	99	1	1
V	75	25	1
W	99	1	1
X	100	0	1
Y	98	2	1
Z	99	1	1
AN	75	25	1
AO	36	64	1
AP	75	25	1
AS	99	1	1
AV	25	75	2
AW	17	83	2
AX	82	18	2
AY	17	83	2
AZ	25	75	2
BA	17	83	2
BB	17	83	2
BC	17	83	2
BE	0	100	2
BF	0	100	2
BH	75	25	2
BI	83	17	2



Test Set

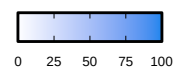
1	7 (70%)	3 (30%)	10 (71%)
2	2 (50%)	2 (50%)	4 (29%)
total	9 (78%)	5 (40%)	14 (64%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set

Probability Heatmap

	1	2	Exp
AA	36	64	1
AB	25	75	1
AJ	100	0	1
AK	82	18	1
G	25	75	1
I	75	25	1
M	75	25	1
AQ	89	11	1
AT	100	0	1
AU	75	25	1
BD	25	75	2
BG	75	25	2
BJ	25	75	2
BK	83	17	2



9th Place

Table:

Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
90.7%	0.653

Model Coefficients

Term	Estimate
(Intercept)	23.362478
NPA_5_P	-66.223045
fr_Ar_N	-43.598682
fr_aryl_methyl	-5.157459
fr_benzene	-5.245496

Training Set

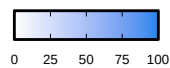
1	30 (97%)	1 (3%)	31 (72%)
2	3 (25%)	9 (75%)	12 (28%)
total	33 (91%)	10 (90%)	43 (91%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	96	4	1
AF	99	1	1
B	97	3	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	95	5	1
AD	95	5	1
AE	100	0	1
AH	92	8	1
AI	100	0	1
AL	100	0	1
AM	83	17	1
H	63	37	1
K	79	21	1
N	100	0	1
O	70	30	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	100	0	1
V	96	4	1
W	100	0	1
X	100	0	1
Y	100	0	1
Z	100	0	1
AN	83	17	1
AO	9	91	1
AP	66	34	1
AS	100	0	1
AV	21	79	2
AW	63	37	2
AX	56	44	2
AY	0	100	2
AZ	0	100	2
BA	21	79	2
BB	1	99	2
BC	11	89	2
BE	0	100	2
BF	0	100	2
BH	78	22	2
BI	26	74	2



Test Set

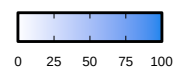
1	7 (70%)	3 (30%)	10 (71%)
2	2 (50%)	2 (50%)	4 (29%)
total	9 (78%)	5 (40%)	14 (64%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set

Probability Heatmap

	1	2	Exp
AA	5	95	1
AB	8	92	1
AJ	100	0	1
AK	94	6	1
G	5	95	1
I	89	11	1
M	99	1	1
AQ	69	31	1
AT	100	0	1
AU	80	20	1
BD	1	99	2
BG	2	98	2
BJ	56	44	2
BK	64	36	2



10th Place

Table:

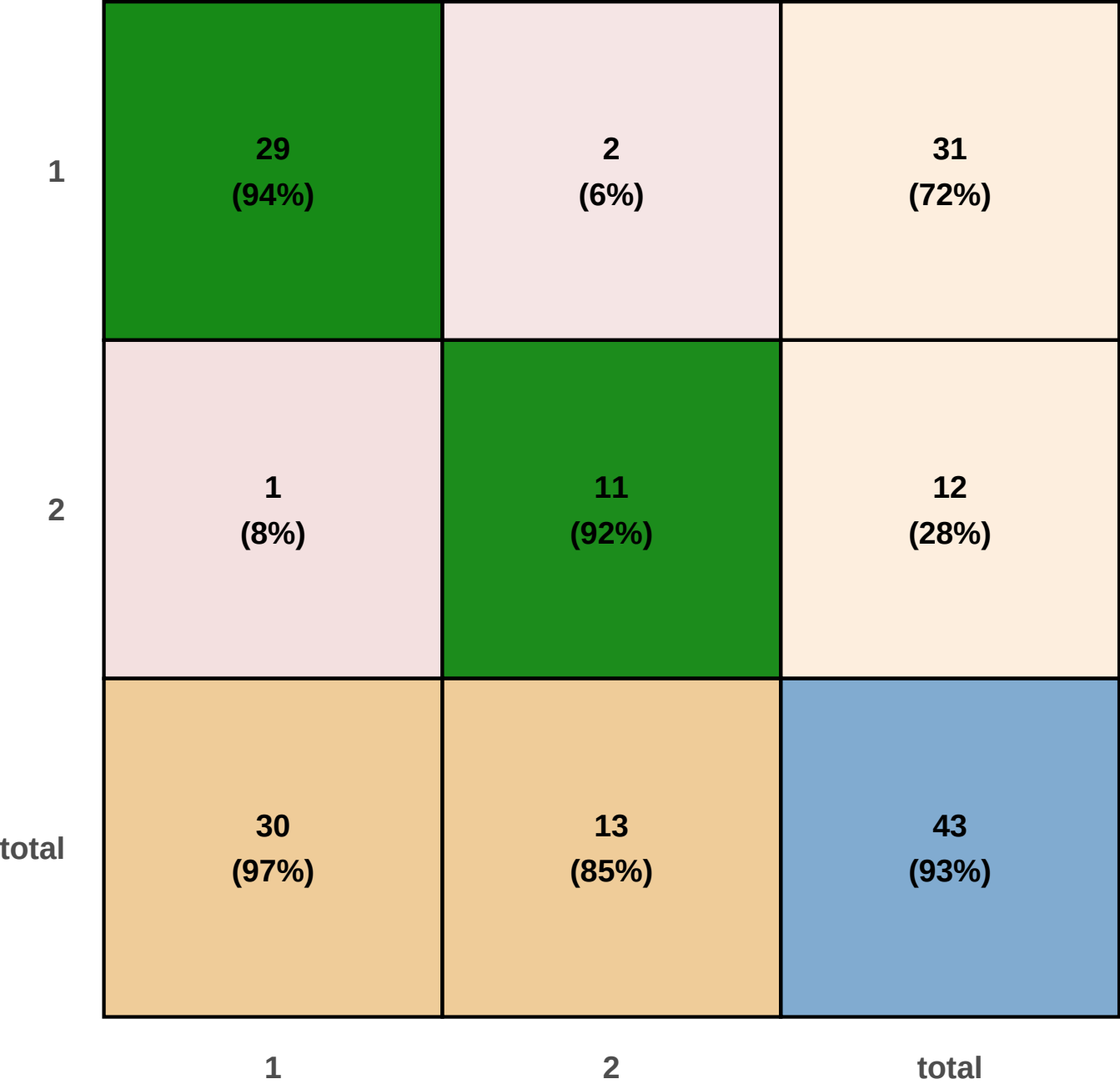
Full Model Stats - Overall Accuracy and Pseudo-R2

Accuracy	McFadden_R2
93.02%	0.644

Model Coefficients

Term	Estimate
:-----:	-----:
(Intercept)	21.571446
NPA_Avg_H_onPhos	-101.577802
fr_Ar_N	-12.283740
fr_aryl_methyl	-3.698223
fr_benzene	-4.750134

Training Set

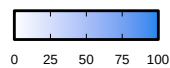


■ Accuracy ■ False ■ Precision ■ Size ■ True

Training Set

Probability Heatmap

	1	2	Exp
A	100	0	1
AF	90	10	1
B	98	2	1
D	100	0	1
E	100	0	1
F	100	0	1
AC	94	6	1
AD	100	0	1
AE	100	0	1
AH	88	12	1
AI	100	0	1
AL	100	0	1
AM	78	22	1
H	19	81	1
K	91	9	1
N	100	0	1
O	97	3	1
P	100	0	1
R	100	0	1
S	100	0	1
T	100	0	1
U	100	0	1
V	93	7	1
W	99	1	1
X	100	0	1
Y	70	30	1
Z	100	0	1
AN	95	5	1
AO	14	86	1
AP	94	6	1
AS	100	0	1
AV	8	92	2
AW	19	81	2
AX	84	16	2
AY	18	82	2
AZ	13	87	2
BA	49	51	2
BB	21	79	2
BC	9	91	2
BE	0	100	2
BF	0	100	2
BH	29	71	2
BI	31	69	2



Test Set

1	7 (70%)	3 (30%)	10 (71%)
2	2 (50%)	2 (50%)	4 (29%)
total	9 (78%)	5 (40%)	14 (64%)
	1	2	total

■ Accuracy ■ False ■ Precision ■ Size ■ True

Test Set
Probability Heatmap

	1	2	Exp
AA	19	81	1
AB	6	94	1
AJ	100	0	1
AK	82	18	1
G	2	98	1
I	94	6	1
M	91	9	1
AQ	89	11	1
AT	100	0	1
AU	94	6	1
BD	14	86	2
BG	91	9	2
BJ	21	79	2
BK	95	5	2

